

CYBER-ATTACK DETECTION USING MACHINE LEARNING

A Research & Development Based Project Report

Submitted in Partial Fulfillment of the Requirement for the Award of Degree of

BACHELOR OF COMPUTER APPLICATIONS



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CERTIFICATE

This is to certify that the Major Project Report entitled “CYBER-ATTACK DETECTION USING MACHINE LEARNING” submitted by Ms. Aman Verma, Roll No. 2023CSCU417, in partial fulfillment for the award of the degree of Bachelor of Computer Applications (BCA) to Sri Guru Granth Sahib World University, Fatehgarh Sahib, is a bona fide record of the work carried out by him under my supervision and guidance during the academic session 2023–2026. At Sri Guru Granth Sahib World University.

To the best of my knowledge, this work has not been submitted elsewhere for the award of any other degree, diploma or certificate.

The project work is found satisfactory and approved for submission.

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CANDIDATE'S DECLARATION

I, Aman Verma, hereby certify that the work which is being presented in the project report entitled "CYBER-ATTACK DETECTION USING MACHINE LEARNING" in fulfillment of the requirements for the award of the Degree of Bachelor of Computer Applications (B.C.A.) submitted to the Department of Computer Science, Sri Guru Granth Sahib World University, Fatehgarh Sahib, is an authentic record of my own work carried out by me during the academic session 2025–2026 under the supervision and guidance of Mr. Ramandeep Singh, Assistant Professor, Department of Computer Science, Sri Guru Granth Sahib World University, Fatehgarh Sahib.

This project has been developed as a part of my Final Year Major Project Submission for the completion of the B.C.A. degree. The work presented in this report is original and has been completed by me with sincere effort, study, analysis, coding, testing, and implementation.

I further declare that the matter embodied in this project report has not been submitted by me earlier for the award of any degree or diploma of this or any other university/institution.

I also certify that all the sources of information used in this project have been duly acknowledged.

This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

Date: _____

Mr. Ramandeep Singh

Project Guide

(Guide Signature: _____)

Aman Verma

B.C.A Final Year Student

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I take this opportunity to express my deep sense of gratitude and sincere thanks to all those who have directly or indirectly helped me in the successful completion of my Final Year Project titled “CYBER-ATTACK DETECTION USING MACHINE LEARNING.”

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ABSTRACT

The rapid growth of internet connectivity and digital infrastructure has led to a significant rise in cyber threats and network-based attacks, creating serious security challenges for individuals, organizations, and institutions. One of the major problems faced by network administrators and security professionals is the inability to monitor and detect malicious network activity in real time, which often results in delayed responses, data breaches, and system compromises. To address this critical issue, the present project titled "Cyber-Attack Detection Using Machine Learning" has been developed as an intelligent real-time network monitoring application that captures live network packets, analyses traffic patterns, and classifies potential cyber threats based on the behaviour of incoming and outgoing data.

The main objective of this project is to design and implement a smart security platform that can continuously monitor network traffic and automatically identify suspicious activity such as Denial of Service attacks, Port Scanning, SQL Injection attempts, Cross-Site Scripting, and Malware communication by analysing packet-level features including source port, destination port, packet size, protocol type, and TCP flag patterns.

This application has been developed using Python as the primary programming language, with Streamlit as the web-based dashboard framework for real-time visualization. PyShark has been used as the Python interface to Wireshark/TShark for live packet capture, while Npcap serves as the Windows kernel-level packet capture driver. SQLite Database has been used for persistent storage of all captured events, risk scores, and detection logs. Python Threading and Queue modules have been implemented to run background capture workers without blocking the user interface, ensuring smooth and uninterrupted real-time monitoring. The system assigns a risk score between 0.0 and 1.0 and a confidence score to every captured packet, generates high-risk alerts for threats exceeding the threshold, and provides a Recommendations Engine that suggests step-by-step corrective actions based on the detected attack type.

The proposed system eliminates the need for manual packet inspection and significantly reduces the response time to network threats by providing continuous automated monitoring in both Live and Background tracking modes. It serves as a practical and accessible security tool for students, cyber security learners, and network administrators who require an affordable and easy-to-deploy threat detection solution.

This final year project not only demonstrates the practical implementation of network security concepts, real-time data processing, and web dashboard development but also highlights the growing importance of automated threat detection systems in modern cyber security. The developed application is user-friendly, efficient, and capable of further enhancement in future with the integration of advanced trained Machine Learning models for higher detection accuracy and broader attack classification.

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